(Please write your Enrollment Number) Enrollment No._____

MID-TERM EXAMINATION
(Course Name: B. Tech.) (Semester: 3rd)
(October, 2025) OFF-LINE mode

Subject Code: BAS 203	Subject Name: Numerical Techniques for Engineers
Time: 1½Hours	Maximum Marks: 30

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3. A scientific, non-programmable calculator is permitted in the exam.

Q1		(2.5*4=10)	CO Mapping
	a) Given two numbers $y_1 = 3.456$ and $y_2 = 27.8$ are corrected up to their significant digits. Compute the relative errors in $y_1 + y_2$ and $y_1 - y_2$.		1
	b) Express the number 54.625 in the following forms: (i) Base 2 floating point representation. (ii) 32-bit IEEE-754 single precision bit string.		1
	c) Suppose $f(x) = x^3 - 7x + 6$. (a) State the condition under which the Newton-Raphson method will converge to a root of this function for a chosen initial value x_0. (b) Verify if the method is likely to converge when $x_0 = 1.5$.		2
	d) Given the system of equations: $\begin{cases} x + y = 2 \\ x + 1.0001y = 2.0001 \end{cases}$ **(i)** Calculate the condition number of the coefficient matrix. **(ii)** If the right-hand side changes to $\begin{bmatrix} 2 \\ 2 \end{bmatrix}$, solve the system again and explain what this tells you about the system's conditioning.		3

	UNIT-I		CO Mapping
Q2	**Attempt any two parts**	(5*2=10)	
	a) Find the root of the equation $\cos x - 5x + 1 = 0$ using the fixed-point method, correct to four decimal places, by taking an initial approximation $x_0 = 0.5$.		2
	b) Find one root of the equation $e^x - 3x = 0$, correct to two decimal places, using the method of bisection.		2
	c) Find the double root of the equation $x^3 - x^2 - x + 1 = 0$.		2

	UNIT-II		CO Mapping
Q2	**Attempt any two parts**	**(5*2=10)**	
	a) Solve the following system of equations by the LU decomposition method: $4x + y - z = -2$ $5x + y + 2z = 4$ $6x + y + z = 6$		3
	b) Solve the following system of equations: $12x_1 + 7x_2 + 3x_3 = 2$ $x_1 + 5x_2 + x_3 = -5$ $2x_1 + 7x_2 - 11x_3 = 6$ by the Gauss-Seidel method. Begin with an initial guess (1,3,5) and perform three iterations.		3
	c) Given a matrix $A = \begin{bmatrix} 4 & 1 & 2 \\ 1 & 3 & 1 \\ 2 & 1 & 3 \end{bmatrix}$, use the power method starting with the initial vector $[1 \quad 1 \quad 1]^T$ to find the dominant eigenvalue and eigenvector after 5 iterations.		3